The Architecture of Opportunity

A Deterministic Data Pipeline for Market-Aware Content Ideation

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What topic could be interesting in this period?

Can someone help me with some ideas?

Which video can I create today?

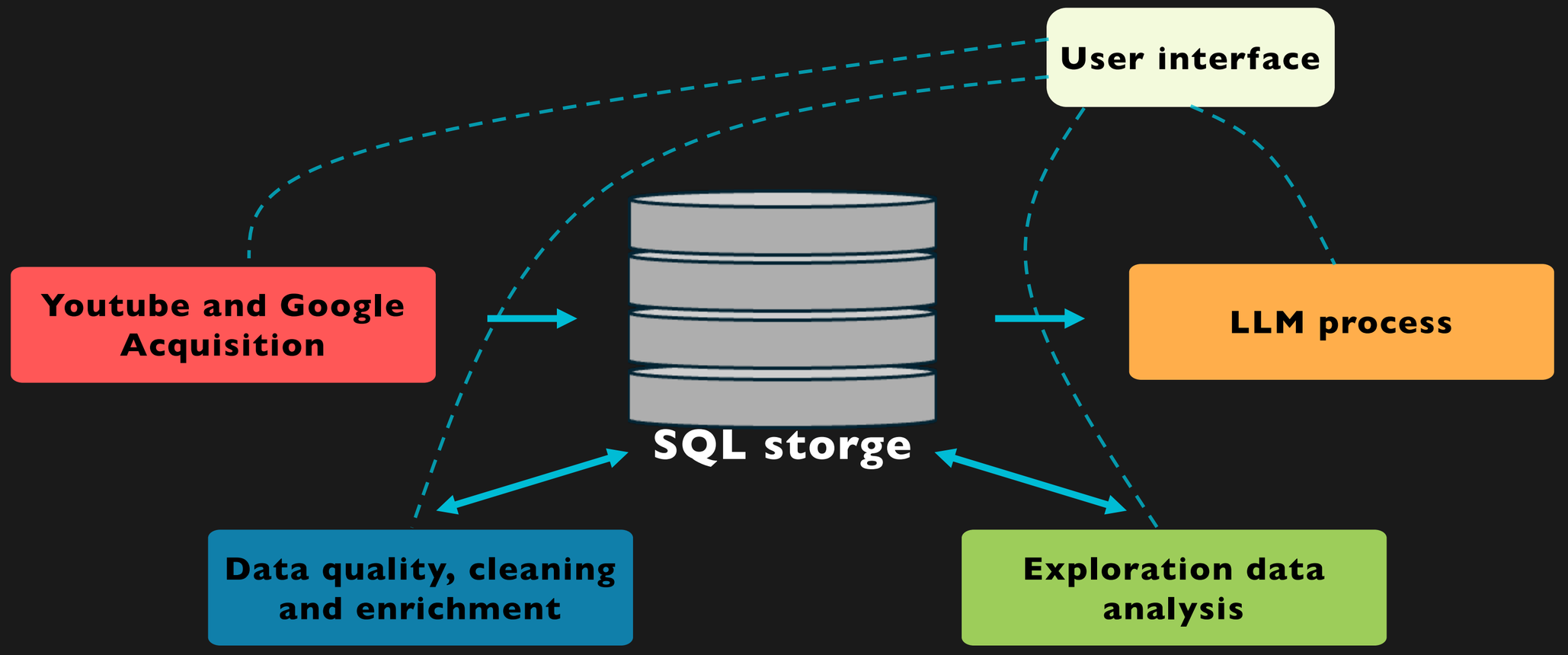
Can we transform heterogeneous web signals into a deterministic, pre-publication decision-support system for content ideation?

Data integration and decision-support problem by independently observing, validating, and analytically combining supply and demand signals.

Design a local, deterministic ideation engine that translates validated YouTube performance and google search intent into actionable pre-publication guidance.

STORAGE ARCHITECTURE & DATA MANAGEMENT DESIGN

Multiple tier architecture



YOUTUBE ACQUISITION & RAW DATA

- **YouTube data API v3**
- **Keyword-based discovery on: Titles, description and tags**
- **Global lookback period split into small, disjoint windows**
- **Results serialized as NDJSON**

YouTube video structure

```
{"Video_id": "string"
 "Title": "string",
 "Description": "string",
 "Published_at": "iso-8601 datetime",
 "Channel": {"id": "string",
             "Title": "string"},
 "Discovery": {"keyword": "string",
               "Keyword_variant": "string",
               "Order": "relevance | date | viewcount",
               "Window": {"start": "iso-8601 datetime",
                          "end":"iso-8601 datetime"},
               "Matched_on": {"title": 0 | 1, "tags": 0 | 1,
                              "description": 0 | 1}},
  "Collected_at": "iso-8601 datetime",
  "Api_key_used": "integer"}
```

YouTube video statistics structure

```
{"Video_id": "string",
 "Statistics": {"view_count": "integer", "Like_count": "integer",
                "Comment_count": "integer",
                "Favorite_count": "integer",
                "Duration_seconds": "integer", "Is_reel": 0 | 1},
 "Collected_at": "iso-8601 datetime",
 "Api_key_used": "integer"}
```

YouTube channel statistics structure

```
{"Channel_id": "string",
 "Subscriber_count": "integer",
 "Video_count": "integer",
 "View_count": "integer",
 "Hidden_subscribers": true | false,
 "Collected_at": "iso-8601 datetime",
 "Api_key_used": "integer"}
```

GOOGLE TRENDS

Data is collected over multiple predefined temporal windows and ingested only through strictly validated file names, ensuring deterministic context reconstruction, explicit temporal structure, and complete protection against silent data contamination.

Top queries gives stable, high-volume demand

query	search_interest	rank	keyword	query_type	window_days	window_start	window_end	logical_name	source_file	ingested_at
TEXT	INTEGER	INTEGER	TEXT	TEXT	INTEGER	TIMESTAMP	TIMESTAMP	TEXT	TEXT	TIMESTAMP

Rising queries shows emerging or accelerating interest

query	increase_percent	rank	keyword	query_type	window_days	window_start	window_end	logical_name	source_file	ingested_at
TEXT	REAL	INTEGER	TEXT	TEXT	INTEGER	TIMESTAMP	TIMESTAMP	TEXT	TEXT	TIMESTAMP

STORAGE

Relational SQL
Impose explicit schemas, typed attributes

ELT, Not ETL
Raw data ingested first, transformations executed natively in SQL

Topic-First Data Isolation
Keyword-prefixed tables enable multi-topic separation, incremental updates

Fail-Safe Acquisition
NDJSON and CSV used as checkpoints, allowing re-runs without re-acquisition

YOUTUBE DATA QUALITY AND ENRICHMENT

Only structurally valid and consistent YouTube records are considered

Exposure efficiently

How efficiently a video converts channel size into views

$$EE_v = \text{exposure_efficiency}_v = \frac{V_v}{S_c}$$

Engagement rate

Captures user interaction intensity per view

$$ER_v = \text{engagement_rate}_v = \frac{L_v + C_v}{V_v}$$

Metrics are percentile-normalized

Best content score

Content that already performs well across multiple dimensions

$$\text{Best_Content_Score}_v = 0.40 \cdot ER_v^p + 0.30 \cdot EE_v^p + 0.30 \cdot V_v^p$$

Opportunity score

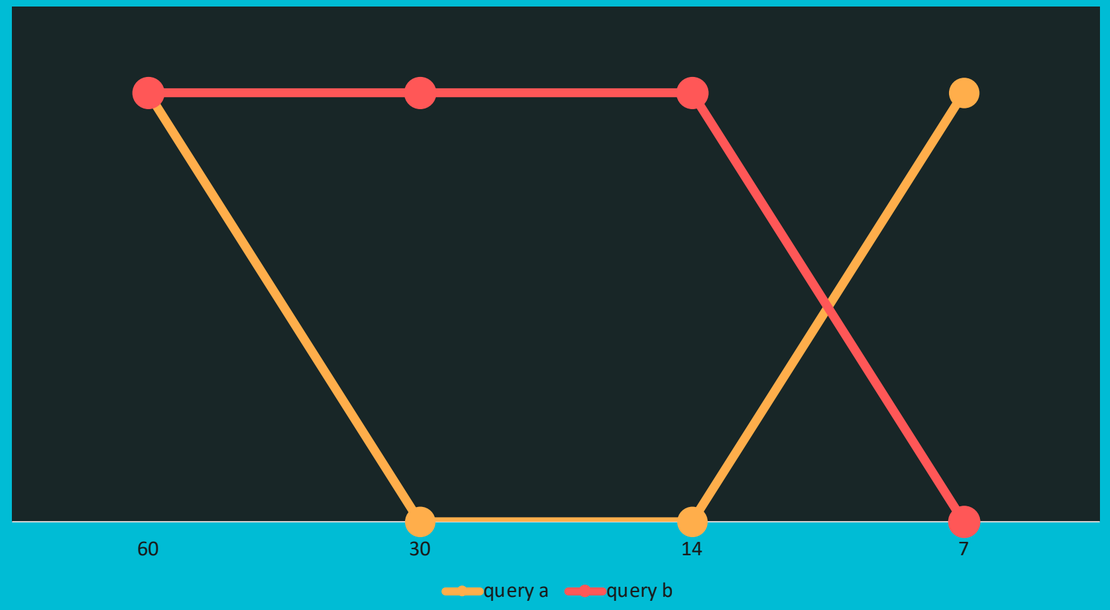
High-engagement content with underexposed potential

$$OS_v = 0.40 \cdot ER_v^p + 0.40 \cdot \left(1 - EE_v^p\right) + 0.20 \cdot \left(1 - V_v^p\right)$$

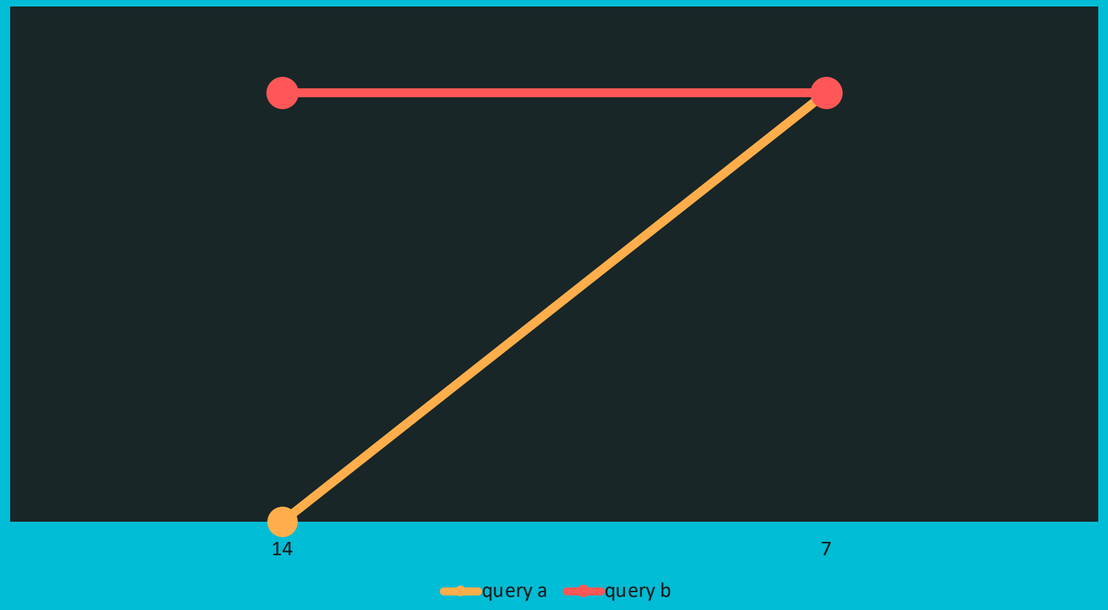
GOOGLE DATA QUALITY AND ENRICHMENT

Raw CSV Google trends → **Normalization «Breakout» →5000% Growth % → float** → **Ranking and Pattern creation**

GOOGLE RISING QUIRIES – PATTERN EVALUATION

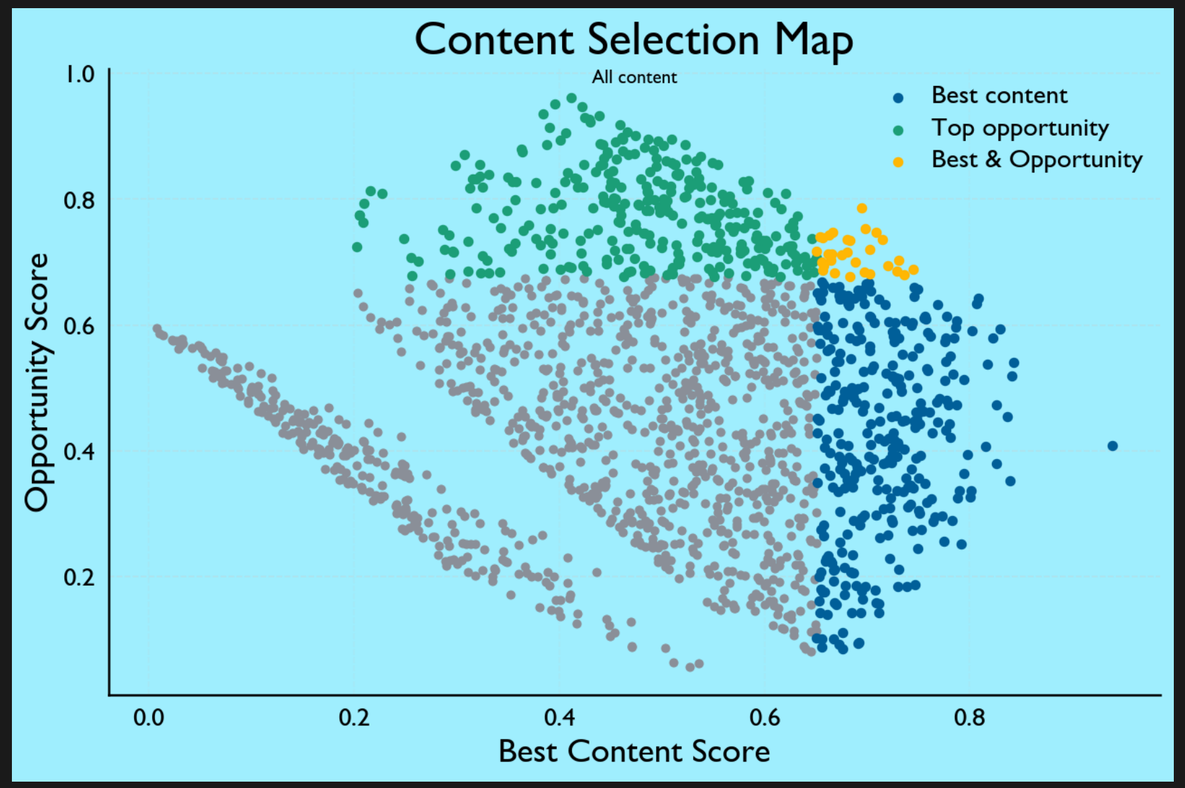


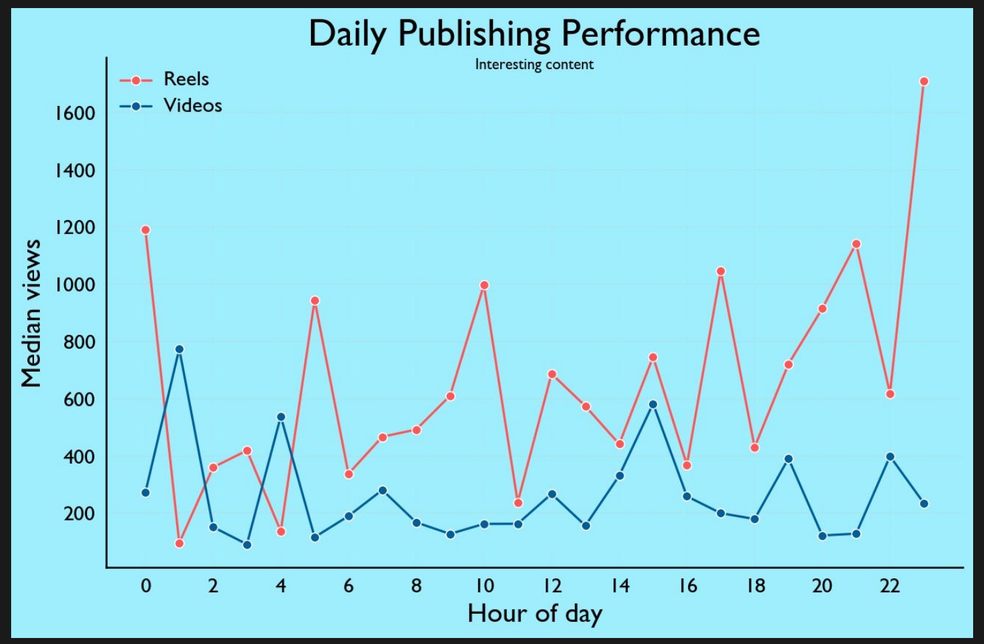
GOOGLE TOP QUERIES – PATTERN EVALUATION

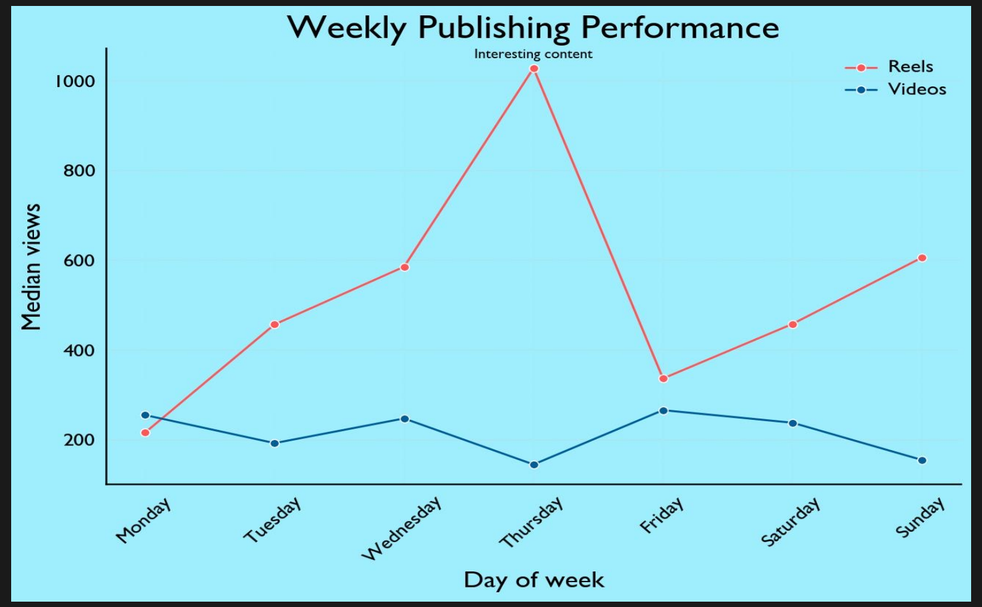


YOUTUBE EDA

- Select 20%
  - Best content score
  - Opportunity score
- Video and reel comparison
- Welch's test







LARGE LANGUAGE MODEL

LLM reasoning

- The input is only structured context and has its own specific prompt template
- Separation between reasoning and decisioning
- Has no access to raw data
- Specific prompt are given to ensure a structured reasoning

Inputs

- Title and description of the selected interesting videos
- Top and rising queries that are interesting
- Publishing days and hours table
- Statistical significance tests

LLM used

Gpt-4.1-mini, configured for constrained and deterministic text synthesis

LLM RESULTS

Preferred content type : VIDEO
Confidence level : medium
Justification : Among the three metrics tested, only comment_count is statistically significant with p_value=0.0041. The diff_pct for comment_count is -57.1%, which favors videos...

VIDEO IDEAS
VIDEO 1
TITLE: How to Crack AI Roles in IT in 2026 | Complete Roadmap & Skills for Data Scientist, ML Engineer, GenAI Engineer
TAGS : AI roles 2026, Data Scientist career, Machine Learning Engineer, GenAI Engineer roadmap, AI job preparation, Data Science career 2026, AI skills 2026, IT AI jobs
KEYWORDS : how to crack AI roles in IT 2026, data scientist career roadmap 2026, machine learning engineer skills 2026, genai engineer job preparation, AI roles in IT 2026, data science career growth 2026, best AI jobs 2026, skills for AI engineer
PUBLISH TIME : Sunday at 01:00
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VIDEO 2
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VIDEO 3
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REEL IDEAS
REEL 1
TITLE: The Struggle of Live Tech Interviews | Data Science Memes #shorts
TAGS: datascience, interview, techmemes, careeradvice, jobinterview, shorts
KEYWORDS: data science interview, tech interview struggles, data science memes, job interview tips, interview preparation
PUBLISH TIME: Sunday at 23:00
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REEL 2
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REEL 3
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LIMITATIONS AND FUTURE WORK

- API keys required → Vertical (Paid) Scalability
- Storage redundancy
- Manual trend ingestion → Enterprise DaaS (DataForSEO)
- Static scoring policies → Dynamic weighting
- Forward-only mode → incorporate post-publication performance feedback

REFERENCES

- YouTube data API documentation: https://developers.Google.Com/youtube/v3
- Google trends terms and limitations: https://policies.Google.Com/
- OpenAI API documentation: https://openai.Com/it-it/api/

- All images and example are took from the run on Febuary 6th 2026

THANK YOU

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